

PROJECT TIMELINE: HOSPITAL MANAGEMENT SYSTEM (HMS)

Project Overview

The **Hospital Management System (HMS)** is a unified, full-stack digital platform designed to streamline healthcare operations, enhance patient care through digitized medical records, and optimize administrative efficiency. Built on a modern tech stack (FastAPI Backend, React Frontend), the system provides real-time visibility into hospital metrics and automated workflows.

Identified Modules & Features

1. Unified Authentication & Identity

- **Centralized Security:** JWT-based authentication for secure session management.
- **Identity Features:** Self-service signup, secure login/logout, and profile management.
- **Access Recovery:** Automated password reset workflow via integrated mail services.

2. Patient & Clinical Core (EMR)

- **Patient Directory:** Comprehensive patient profile management (Registry).

- **Electronic Medical Records (EMR):** Digitized history of patient visits and clinical notes.
- **Appointment Lifecycle:** Scheduling, tracking, and approval workflow (Integrated with n8n Webhooks).

3. Ancillary Services & Ancillary Units

- **The Lab (Diagnostic Tests):** Management of lab orders, reports, and real-time processing status.
- **The Pharmacy (Pharmacy module):** Stock management, dispensing tracking, and medicine inventory.
- **Emergency Blood Bank:** Tracking real-time blood unit availability and cross-matching activities.

4. Facility & Operational Management

- **Bed Management:** Real-time bed occupancy status across ICU and General Wards.
- **Human Capital (Staff Management):** Directory of medical personnel, shift tracking, and department assignments.
- **Inventory Control:** Global tracking of non-medical facility supplies.

5. Administrative & Financial Intelligence

- **Revenue Cycle (Billing):** Automated invoice generation and payment tracking.
 - **Analytics Dashboard:** High-level metrics for admissions, critical alerts, and periodic revenue.
 - **Reporting Engine:** Specialized data exports for administrative auditing.
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Project Development Log & Roadmap

Weeks 1 - 2: Foundation & Backend Scaffold (March 10 - March 23, 2026)

Status: COMPLETED

- **Kickoff (Mar 10):** Initiated the Unified Hospital Management System project.
- **Core Setup:** Established the FastAPI backend structure and React frontend boilerplate using Vite.
- **Data Modeling:** Defined initial schemas for Users and Patients. Integrated the first iteration of the Analytics Dashboard (Mar 18).

Weeks 3 - 4: Clinical Core UI (March 24 - April 6, 2026)

Status: COMPLETED

- **Feature Sprint 1:** Designed and implemented the core clinical modules.
- **Deliverables:** Completed frontend layouts for **Beds Management**, **Lab (Diagnostics)**, **EMR (Electronic Records)**, and **Global Inventory** (Apr 4).
- **Reporting:** Initiated the Administrative Reporting engine (Reports.jsx).

Weeks 5 - 6: Operational Modules & Security (April 7 - April 19, 2026)

Status: COMPLETED

- **Feature Sprint 2:** Expansion of facility and administrative modules.
- **Deliverables:** Finalized **Staff Directory**, **Pharmacy (Medicines)**, **Revenue Cycle (Billing)**, and **Blood Bank** modules (Apr 10).
- **Security Hardening:** Implemented robust JWT authentication workflows, including Password Recovery (Apr 9).

- **Appointment Logic:** Completed the backend approval/denial workflow and added custom fields (Telegram integration) for status notifications (Apr 17).

Week 7: Automation & Verification (April 20 - April 26, 2026)

Status: *IN PROGRESS*

- **Current Milestone:** Integrated the **n8n Webhook notification system** for automated patient alerts.
- **Quality Assurance:** Executed a comprehensive API test suite covering all 10+ modules with 100% success (Apr 20).
- **Documentation:** Finalized professional project documentation and generated evidentiary PDF reports for Team Lead review.

Weeks 8 - 12: Production Hardening & Global Launch (Forecasted)

- **Phase Goal:** Migration from SQLite to PostgreSQL production environment.
- Implementation of automated CI/CD pipelines and load testing.
- Final user acceptance testing (UAT) and high-fidelity reporting dashboard optimization.
- **Final Go-Live:** Scheduled for Week 12.

Key Deliverables & Actual Milestones

Milestone	Date	Deliverable
M1: Alpha Baseline	Mar 18	Functional Dashboard and Core Data Models.
M2: Clinical MVP	Apr 4	Clinical Layouts (EMR, Lab, Beds) functional.

Milestone	Date	Deliverable
M3: Module Expansion	Apr 10	Pharmacy, Billing, and Blood Bank integration.
M4: Beta Freeze	Apr 20	Webhook automation and 100% verified API suite.
M5: Production Launch	(Week 12)	Final Multi-platform Deployment.